Pipped at the Post: Pipecolic Acid Derivative Identified as SAR Regulator

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The non-protein amino acid pipecolic acid (Pip) is a lysine catabolite involved in plant systemic acquired resistance (SAR). In this issue of *Cell*, Hartmann et al. (2018) demonstrate that a flavin-dependent monooxygenase converts Pip to N-hydroxypipecolic acid (NHP), which functions as a critical metabolic regulator of SAR in *Arabidopsis*.

Plants and animals live in an environment teeming with potentially harmful microbes and have therefore evolved immune systems to defend against infections. In plants, local induction of immunity upon infection often triggers resistance to subsequent pathogen attacks in distal tissues, a phenomenon called systemic acquired resistance (SAR) (Fu and Dong, 2013). Accumulation of the plant hormone salicylic acid (SA) is a hallmark of SAR and is generally considered the primary driving force for the systemic upregulation of defense-related genes (Fu and Dong, 2013). However, several mutations causing SAR deficiency have been found in genes responsible for the biosynthesis of a lysine (Lys) catabolite known as pipecolic acid (Pip) (Ding et al., 2016; Návarová et al., 2012). Two of these genes, AGD2-like defense response protein 1 (*ALD1*) encoding a Lys aminotransferase and SAR-deficient 4 (*SARD4*) encoding a homolog of bacterial ornithine cyclodeaminase, are essential for both the local and systemic accumulation of Pip. Exogenous application of Pip is able to rescue SAR deficiency in these mutants, implicating Pip as an important regulator of SAR (Ding et al., 2016; Návarová et al., 2012). The biosynthesis of Pip involves ALD1-mediated transamination, cyclization, and isomerization of L-Lys to 2,3-dehydropipecolic acid (2,3-DP) followed by subsequent reduction to Pip by the reductase SARD4 (Ding et al., 2016; Hartmann et al., 2017). One previously enigmatic observation, however, was that Pip-induced SAR depends on a functional flavin-dependent monooxygenase 1 (*FMO1*) gene, mutation of which does not dramatically affect Pip accumulation upon infection (Bernsdorff et al., 2016; Ding et al., 2016). In this issue of *Cell*, Hartmann et al. provide compelling evidence that FMO1 is responsible for the hydroxylation of Pip to N-hydroxypipecolic acid (NHP), the downstream metabolic mediator of what was previously considered Pip-induced SAR (Figure 1) (Hartmann et al., 2018).

Experiments conducted by Hartmann et al. confirm that Pip is a key player in the establishment of SAR but, more importantly, that the *fmo1* mutant is almost completely insensitive to exogenous Pip application. This supports a major role of FMO1 in the Pip-triggered response. A series of state-of-the-art gas chromatography-mass spectrometry (GC-MS) and gas chromatography-fourier transform infrared spectroscopy (GC-FTIR) metabolite analyses identified NHP as the N-hydroxy derivative of Pip, which was further confirmed by an independent chemical synthesis, *in vitro* enzymatic assay with FAD^+ and NADH as co-factors, and *in vivo* feeding assay with Lys isotopic variants. Thus, FMO1 is an N-hydroxylating monooxygenase that generates a plant-derived N-hydroxylated amino acid derivative, NHP.

Further clarification of the SAR pathway was accomplished using a set of comprehensive genetic and metabolic complementation experiments using different metabolite treatments in various genetic backgrounds. Consistent with the role of ALD1 in biosynthesis of Pip and FMO1 in converting Pip to NHP, pretreatment with Pip or NHP increased plant resistance to both bacterial and oomycete pathogens in the *ald1* mutant as well as in wild-type plants. The *fmo1* mutant, however, is defective in Pip-mediated resistance but capable of NHP-mediated resistance to both pathogens. Notably, pretreatment with NHP rendered plants almost symptomless upon both bacterial and oomycete infections in wild-type and *ald1* and *fmo1* mutants, indicating the potential utility of NHP in agricultural applications to improve plant resilience to a broad-spectrum of pathogens.

SA is another major molecule important for mediating SAR (Fu and Dong 2013). It has been shown before that Pip-triggered SAR is both SA dependent and SA independent, and SA may function as an amplifier for Pip-triggered responses (Bernsdorff et al., 2016). The results from this study further substantiate this conclusion. The SA biosynthesis mutant SA induction deficient 2 (*sid2*) and signaling mutant non-expresser of PR gene 1 (*npr1*) did not compromise pathogen-induced Pip or NHP accumulation, but Pip- or NHP-mediated resistance requires an intact SA biosynthesis pathway. Surprisingly, NHP, but not Pip, accumulated to a higher level in both *sid2* and *npr1* mutants. The authors suggest that SA exhibits a negative regulatory role on the conversion of Pip to NHP (Figure 1). Because of the structural resemblance of NHP and SA with respect to similar hydroxyl and carboxyl groups, an alternative

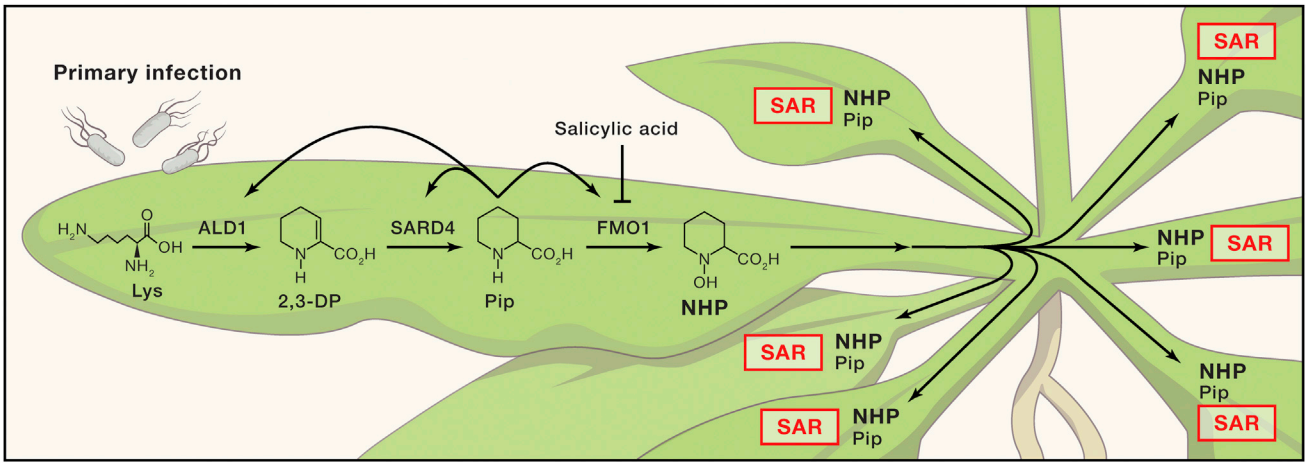


Figure 1. N-hydroxypipecolic Acid Induces Plant Systemic Acquired Resistance

In plants, local pathogen infection triggers systemic acquired resistance (SAR) in distal tissues protecting from subsequent pathogen attacks. Upon infection, N-hydroxypipecolic acid (NHP) is generated by a three-step biochemical process: the Lys aminotransferase ALD1 converts L-Lys to 2,3-dehydropipecolic acid (2,3-DP), followed by a subsequent reduction to pipecolic acid (Pip) by the reductase SARD4. Pip is then N-hydroxylated by the monooxygenase FMO1 to generate NHP, a mediator of SAR. Both Pip and NHP accumulate in locally infected and distal leaves upon infection. Pip accumulation transcriptionally upregulates the expression of NHP biosynthesis genes, *ALD1*, *SARD4*, and *FMO1*. The plant hormone salicylic acid (SA), another key regulator of SAR, negatively regulates the conversion of Pip to NHP.

possibility is that SA might act as a competitive inhibitor of FMO1. The mode of action of acetylsalicylate (aspirin) provides some precedent for this hypothesis based on its activity as an inhibitor of an oxygenase in mammalian systems (Vane and Botting, 2003). Significantly, Pip treatment also increased expression of *ALD1*, *SARD4*, and *FMO1*, suggesting a positive feedback loop of Pip/NHP biosynthesis in regulating SAR (Figure 1).

The findings from this study may aid in addressing key questions in the mechanistic understanding of SAR. To activate SAR, a mobile signal must be generated in the locally inoculated tissues and then transported systemically to the distal tissues via the vasculature (Figure 1) (Dempsey and Klessig 2012, Shah et al., 2014). This has led to the identification of several candidate long-distance signals (e.g., methyl SA, dicarboxylic acid azelaic acid, glycerol-3-phosphate-dependent factor, and Pip), some of which may work cooperatively for proper SAR induction. The results of Hartmann et al. indicate that NHP is the actual regulator of SAR, rather than Pip. Indeed, the authors found that systemic NHP accumulation occurs prior to systemic SA and Pip accumulation. It is possible that NHP is transported from local to distal tissues, which occurs at the very onset of SAR. Future investigations of the role of NHP in long-distance communication may offer insight into the relationship of NHP with other mobile signals. Together, this work represents an important step forward in the elucidation of SAR signaling dynamics and has opened several new lines of inquiry in the field of plant immunity. Ideally, identification of the conversion of Pip to NHP could provide plant breeders with new markers in genetically and metabolically guided screens for crop disease resistance against a broad-spectrum of pathogens, paving the way for improved genetic engineering of resilient agricultural strains.

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